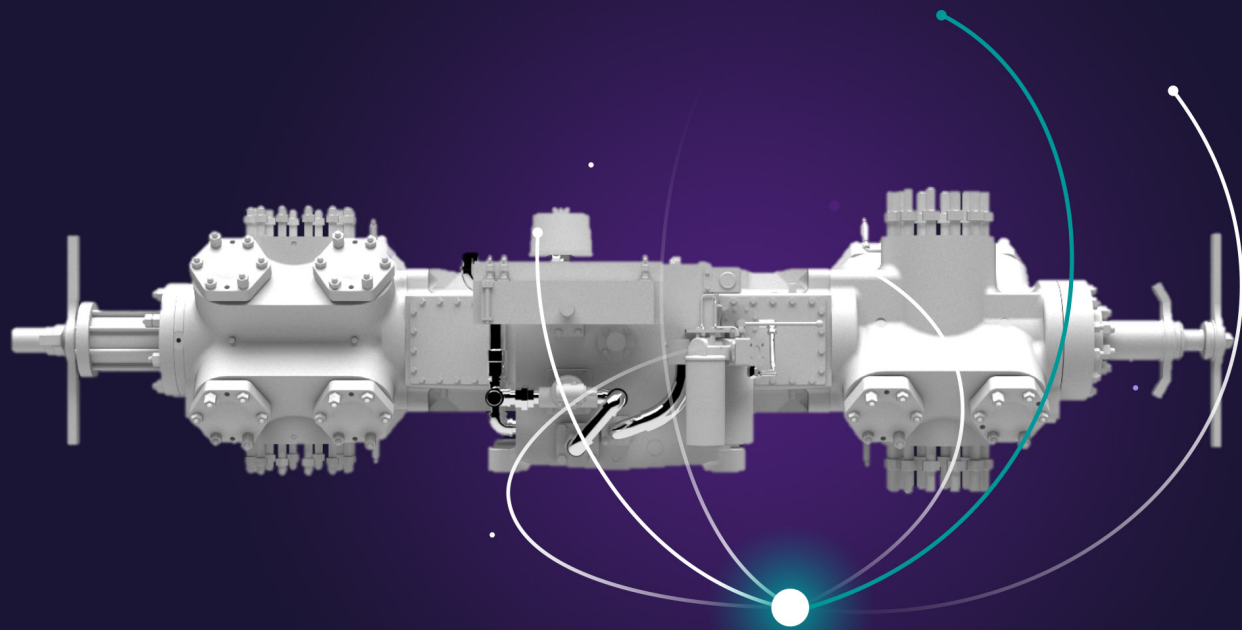


MOS product description



Frame	Multi-throw, balanced-opposed rigid cast iron frame heavily ribbed and reinforced with integrally cast crosshead extensions. Open frame top construction with steel tie rods and individually fitted cast iron spacers over each main bearing. Gasketed doors in top and sides permit easy access to running gear. Inside painted with oil proof white paint. Lower part of frame forms sump for lubricating oil.
Options	• Explosion relief device (Bicera Relief Valve) to prevent over pressurization of crankcase. • Frame liquid level controller to maintain proper oil level. • Immersion oil heaters maintain the oil at proper temperature in colder environments. Watt density is 15W/in². • Dry sump configuration for compressors installed on floating platforms or ships. • Lifting lugs for ease in transportation and installation. Lifting lugs are provided as standard on export shipped compressors.
Frame main bearing	Horizontally split, precision steel backed copper tin shell bearing with babbitt overlay.
Option	• Temperature detection system (RTD) for main bearings, including tubing fittings, brackets and bolting. Terminal connection at frame on outside of the frame.
Crankshaft	Forged, alloy-steel crankshaft with internal passages for pressurized bearing lubrication. Crankshafts are a one-piece design with a straight shaft on the drive end. Two and four throw have counterweights to reduce horizontal forces and moments.
Crankpin bearing	Horizontally split, precision steel backed copper tin shell bearing with a babbitt overlay.
Connecting rod	Forged alloy steel connecting rods are two-piece construction that are held in place by dowel pins and assembled with forged ferry head capscrews with rolled threads. The connecting rod has internal passages to allow pressurized oil lubrication to the crosshead pin and shoes.
Connecting rod bushing	Single piece design made from a solid steel backed bronze bushing that is interference fit and shrunk into place.
Crosshead	Nodular iron box type crossheads with babbbitted running surfaces. Crosshead pin is a semi-floating hardened steel secured in place by cap pin and bolt. Where required, positive-lock disc-type weights are bolted to crosshead for balancing within two pounds maximum.
Oil pump	Oil pump is a positive displacement gear type pump that is externally mounted on the frame end cover and is driven off the end of the crankshaft via a chain and sprocket arrangement. The pump is complete with a pressure relief valve for cold start protection along with an externally adjustable main bearing header pressure regulating valve.
Oil filter	Oil filter is a frame mounted, full flow filter that uses 10 micron spin on canister type disposable elements. The filter housing comes with a 0-30 psig liquid filled differential pressure gauge. External oil piping is stainless steel.
Option	• Dual oil filter with transfer valve shipped separately from the compressor which enables the oil filter to be changed without shutting down the unit.
Oil cooler	Fixed bundle shell and tube water cooled oil cooler shipped separately from the compressor to keep the oil temperature within operating limits.
Distance piece	No distance piece required. Cylinders are close coupled to the frame.
Options	• Cast iron one-piece design, 13.00" long (18.00" long on high pressure with MAWP above 2000 psig), sealed to 25 psig. Gasketed solid covers on both sides. Stainless steel fittings with stainless steel braided hose for lubrication and vent. Inside painted with oil proof white primer paint. Cast iron wiper rings and Teflon sealing rings. Distance piece has provisions for purge. • Cast iron two-piece design, 21.00" long sealed to 25 psig. Consists of one 8.00" distance piece on frame end and one 13.00" distance piece on cylinder end. Partition packing is non-lubricated. Partition packing has an option of being purged.
Cylinder Ø4.75" - Ø20.50"	Single piece non-jacketed cylinder casting design with integral frame head and separate removable outer head. The valves located tangential to the bore in cylinder block and the inlet and discharge directly opposed with top suction and bottom discharge. Top inlet and bottom discharge gas flanges are machined to conform to the dimensional requirements of ANSI B16.5. Cylinders are equipped with two ½" NPT indicator holes per end to monitor cylinder performance.
Options	• Liners are available for all cylinder sizes. • Cylinders can be provided in non-lubricated configuration for 5.00" and 6.00" stroke ranges.

Compressor valves	Siemens Energy Magnum HammerHead Valves. Consists of valve seat, stop plate, elements & springs. Non-metallic (PEEK) valve plate. All cylinders have barrel type crabs and O-ring sealed valve covers. All assemblies are polarized so that an inlet valve cannot be installed in a discharge port.
Options	• PF Valves (plate style) valves that utilize a singular plate sealing element. • Magnum Valves utilize individual poppet elements for low mole weight applications. • Magnum Plus Valves are higher efficiency valves with individual poppet elements and field rebuildable stopplates.
Capacity control and unloading	Each cylinder is supplied with a solid outer head that provides the least amount of added clearance to the cylinder to maximize efficiency.
Options	• A manually operated outer end variable volume clearance pocket (VVCP) can be supplied to easily adjust the capacity of the cylinder. • Solid outer heads with fixed added clearance can be supplied to achieve certain performance conditions if adjustability is not required. • Valve Spacers add fixed clearance to the cylinder to reduce capacity. • Finger Unloaders are utilized with inlet valves to achieve required load steps while minimizing added cylinder clearance. • Plug Unloaders are utilized with partial inlet valves to achieve required load steps. • Port Unloaders displace inlet valves for dedicated unloading location to achieve required load steps. At least one traditional inlet valve per end must be utilized. • Automatic fixed volume clearance pocket (AFVCP) utilizes a port unloader on the outer head of the cylinder for one discrete load step. • Infinite Step Control (ISC) system is a hydraulically controlled finger unloader that allows for a range (30% - 100%) of normal capacity.
Piston	Pistons are designed to utilize a combination piston ring and rider band. The piston is fastened to the piston rod with a Supernut.
Option	• Pistons can be supplied with traditional piston ring and rider bands.
Piston rod	Single piece alloy steel rod with rolled threads.
Options	• Siemens Energy TC3 coated alloy steel rod with rolled threads. • Siemens Energy TC3 coated 17-4 PH stainless steel rod with rolled threads.
Pressure packing	Segmented cups with individual rings fixed to the cylinder by a bolted flange. Cups are fixed together with tie rods to aid in easy installation. Packing rings are multi-piece design and can be installed without removing the piston and rod assembly. Cases are lubricated and vented as standard.
Options	• Packing case purge cup to mitigate gas leakages into the distance piece and frame to prevent damage and limit safety hazards. • Water coolant passages to reduce temperatures in the packing case to increase ring life. • Non-lubricated packing cases for applications that cannot tolerate oil in the process. • RTDs in the packing case to monitor case temperatures to ensure longevity of the packing ring segments.
Cylinder lubrication	Mechanical force feed lubricator driven off crankshaft with individual pumping units manifolded together and fed into divider block system. The pump manifold is equipped with a rupture disc to ensure proper functionality of the system. The divider blocks equipped with visual cycle pin indicators and check valves on every and proximity switch to monitor cycle time of lubricator. All of the lubrication tubing and fittings are made from stainless steel.
Options	• Pump to point system. Individual pumping units feed oil directly into distribution point on the cylinder and packing. • Digital no flow timer (DNFT) to replace proximity switch on primary divider block to monitor lubricator cycle times. • Suction port oilers can be added to the lubrication system to inject atomized oil into the inlet bottle flange to reduce corrosion of the cylinder and valves on corrosive gas applications. • Electric driven lubricator can be supplied on off-mounted panels. • Reset pin indicators can be provided on all divider block elements to aid in identifying a blockage in a tubing line or a distribution point.

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